Introduction to Classification Metrics

Evaluation metrics measure how well a classification model performs. Different metrics are used based on the problem type (balanced or imbalanced dataset) and business requirements.

2. Confusion Matrix

A confusion matrix provides a summary of the model's performance by comparing predicted vs. actual values.

Actual \ Predicted	Positive (1)	Negative (0)
Positive (1)	True Positive (TP)	False Negative (FN)
Negative (0)	False Positive (FP)	True Negative (TN)

- **True Positive (TP)** – Model correctly predicts positive class.
- **False Positive (FP)** – Model incorrectly predicts positive when it is negative (Type I Error).
- **False Negative (FN)** – Model incorrectly predicts negative when it is positive (Type II Error).
- **True Negative (TN)** – Model correctly predicts negative class.

Python Implementation

```
from sklearn.metrics import confusion_matrix

y_true = [1, 0, 1, 1, 0, 1, 0, 0, 1, 0]
y_pred = [1, 0, 1, 0, 0, 1, 0, 1, 1, 0]

cm = confusion_matrix(y_true, y_pred)
print(cm)
```

3. Accuracy & F1 Score

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

- Works well when the dataset is **balanced**.
- Not useful when the dataset is **imbalanced** (e.g., detecting fraud cases where positive cases are rare).

4. F1- Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- **F1 Score balances Precision and Recall**.
- Suitable for **imbalanced datasets**.

Python Implementation

```
from sklearn.metrics import accuracy_score, f1_score

accuracy = accuracy_score(y_true, y_pred)
f1 = f1_score(y_true, y_pred)

print("Accuracy:", accuracy)
print("F1 Score:", f1)
```

4. Precision & Recall

Precision (Positive Predictive Value)

$$Precision = \frac{TP}{TP + FP}$$

- Measures **how many predicted positives are actually positive**.
- High precision = **few false positives**.
- Used when **False Positives are costly** (e.g., medical diagnoses).

Recall (Sensitivity or True Positive Rate)

$$Recall = \frac{TP}{TP + FN}$$

- Measures how many actual positives are detected.
- High recall = few false negatives.
- Used when False Negatives are costly (e.g., fraud detection**).**

Python Implementation

```python
from sklearn.metrics import precision_score, recall_score

precision = precision_score(y_true, y_pred)
recall = recall_score(y_true, y_pred)

print("Precision:", precision)
print("Recall:", recall)
```

5. Sensitivity & Specificity

Sensitivity (Same as Recall)

$$Sensitivity = \frac{TP}{TP + FN}$$

Specificity (True Negative Rate)

$$Specificity = \frac{TN}{TN + FP}$$

- Measures how well the model identifies negative cases correctly.
- High specificity = low false positives.
- Important in disease detection models where False Positives lead to unnecessary treatments.

Python Implementation

```python
specificity = TN / (TN + FP)  # Need to extract TN, FP from confusion matrix
print("Specificity:", specificity)
```

6. True Positive Rate & False Positive Rate

True Positive Rate (TPR) (Same as Recall)

$$TPR = \frac{TP}{TP + FN}$$

False Positive Rate (FPR)

$$FPR = \frac{FP}{FP + TN}$$

- FPR measures how many actual negatives are incorrectly classified as positives.
- A **lower FPR** means fewer **false alarms**.

7. ROC & ROC-AUC

ROC Curve (Receiver Operating Characteristic Curve)

- Plots **TPR vs. FPR** at different classification thresholds.
- A model closer to the top-left corner is **better**.

AUC-ROC (Area Under the Curve - ROC)

- **AUC (0.5)** → Random guessing.
- **AUC (0.7-0.8)** → Good model.
- **AUC (0.9-1.0)** → Excellent model.

Python Implementation

```python
from sklearn.metrics import roc_curve, roc_auc_score
import matplotlib.pyplot as plt

fpr, tpr, _ = roc_curve(y_true, y_pred)
auc_score = roc_auc_score(y_true, y_pred)

plt.plot(fpr, tpr, label=f"AUC = {auc_score:.2f}")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
```

8. Interview Questions & Answers

Q1: What is the best metric for imbalanced classification problems?

A: F1 Score, Precision-Recall Curve, and AUC-ROC.

Q2: What is the trade-off between Precision and Recall?

A: Increasing precision reduces recall and vice versa.

Q3: What does an AUC score of 0.5 mean?

A: The model performs no better than random guessing.

Q4: Why is Accuracy not a good metric for imbalanced datasets?

A: Because it ignores False Positives and False Negatives.

Q5: When should you use Specificity?

A: When detecting negatives is as important as positives (e.g., disease detection).

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9. Important Links & Resources

📎 **Scikit-Learn Metrics Documentation**: https://scikit-learn.org/stable/modules/model_evaluation.html
📎 **Kaggle Classification Metrics Discussions**: https://www.kaggle.com/discussions/getting-started/176415
📎 **Towards Data Science - Classification Metrics**: https://towardsdatascience.com/metrics-to-evaluate-your-machine-learning-algorithm-f10ba6e38234
📎 **Machine Learning Mastery - Model Evaluation**: https://machinelearningmastery.com/classification-accuracy-is-not-enough-more-performance-measures-you-can-use/